

Lighting sensor specifications

Introduction

The Lighting sensor uses a LDR to detect the intensity of the light. The resistance of a light dependent resistor decreases when the intensity of the light increases. The output of the signal is an analog value, the brighter than the light is, the larger the value.

Specifications

- Voltage: 3.3 ~ 5V
- Response time: 20-30ms
- Operating temperature range: -40 ~ +125 °C
- Peak wavelength: 50 nm.
- **Note: better to use 3.3 V at the input.**

How the Photoresistor is working.

Depending on what are you reading, the formula to read the lux is the following:

$$lux = \frac{((V_{ref} * luxRel) * V_{out}) - luxRel}{Rl}$$

Parameter	Value
ADCres	1.0 if you are using read() or 65535 if you are using read_u16()
RI	10KΩ
VRef	5.0 V or 3.3 V
luxRel	500

To give you a reference about the valid lux values:

Light Condition	Illuminance level (lux)
Pitch Dark	0.4-5
Very Dark	6-50
Dark Indoors	51-95
Corridor areas/Toilets	140-145
Rest room	155-195
Entrances/Exits/Canteens	196-295
School Assembly Halls	296-325
Reading	326-355